

13.9 Subtracting Linear Expressions with Rational Coefficients

Lesson Introduction

 Benchmark	MA.7.AR.1.1 Apply properties of operations to add and subtract linear expressions with rational coefficients.		 Learning Targets	- I can subtract linear expressions with rational coefficients. - I can apply properties of operations to add and subtract linear expressions with rational coefficients.	
 Benchmark Coherence	Building On MA.6.AR.1.4 Apply the properties of operations to generate equivalent algebraic expressions with integer coefficients.			Working Toward MA.8.AR.1.2 Apply properties of operations to multiply two linear expressions with rational coefficients.	
 Essential Questions	- What is different about subtracting linear expressions from adding linear expressions? - How do you subtract linear expressions? - How can you apply what you have learned about combining two linear expressions to combining three or more?				
 Lesson Narrative	Students extend their understanding of combining like terms from the previous lesson by exploring four different methods or strategies of subtracting linear expressions. Students analyze examples that start with integer coefficients and constants and build rational constants and coefficients in subtraction expressions with the fewest possible terms.				
 Component Summary	13.9.1 Warm-Up (~5 min.)	Bell work adding linear expressions (<i>SE p. 308</i>)	13.9.5 Guided Practice (10 min.)	Students work with a partner to find the differences of linear expressions (<i>SE p. 311</i>)	
	13.9.2 Exploration (5 min.)	Students use algebra tiles to find the difference of linear expressions (<i>SE p. 308</i>)	13.9.6 Collaboration (10 min.)	Error analysis for analyzing student work on operations with expressions (<i>SE p. 312</i>)	
	13.9.3 Collaboration (10 min.)	Students collaborate to analyze student work using the distributive property and then relate this to algebra tiles (<i>SE p. 309</i>)	13.9.7 Your Turn! (5 min.)	Combine like terms to simplify three linear expressions with rational coefficients (<i>SE p. 313</i>)	
	13.9.4 Guided Instruction (5-10 min.)	Teacher models how to find the differences of linear expressions by adding the opposite or distributing a negative (<i>SE p. 310</i>)	13.9.8 Wrap-Up (~5 min.)	Reflection on a mistake from today's lesson (<i>SE p. 313</i>)	
 Resources	Glossary		Materials		Additional Preparation
	<u>Key Terms</u> N/A <u>Additional Terms:</u> - Additive inverse - Distributive property		- Algebra tiles (Optional) Notecards (Optional) Chart paper (Optional) Markers		- If using concrete algebra tiles, determine how many tiles are needed to model all the examples in the lesson and organize student sets in a container or ziplock bag. - Determine which students will work together for 13.9.6 Collaboration.

 Teacher Tips	Common Misconceptions - Students may forget to distribute subtraction, or (-1), to all terms in the parentheses (like in the 13.9.2 Exploration). - Students may not simplify the expression completely or have trouble combining all of the like terms in the expressions.	Things to Consider - When adding decimals, consider providing a model or anchor chart for rewriting problems vertically and line up the decimal places. When adding fractions, students may need to be reminded to obtain a common denominator. - If time is an issue, then consider assigning 13.9.6 Collaboration and/or 13.9.7 Your Turn! for homework.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Notice and Wonder 13.9.4 Guided Instruction provides an opportunity for students to Notice and Wonder about two methods for subtracting linear expressions. Students complete a table using the additive inverse method and applying the distributive property method. Engage the class in a Notice and Wonder routine to draw conclusions and connections between the methods.	MA.K12.MTR.2.1: Multiple Representations In 13.9.4 Guided Instruction, students apply two different methods to subtract linear equations. Use this as an opportunity for students to make connections between the two methods and understand how to solve linear subtraction in multiple ways.
 Differentiate Instruction	Consider providing Math Nation’s On-Ramp video for operations with fractions as scaffolding for students when operating with mixed numbers or fractions greater than one (as seen in 13.9.5 Guided Practice and 13.9.7 Your Turn!). Avoid a previous learning target becoming a barrier to conceptual understanding for subtracting expressions by providing scaffolding.	
Lesson Notes:		

13.9.1 Warm-Up: Bell Work

Component Overview: Students combine like terms in algebraic expressions to find their sums.



13.9.1 Warm-Up: Bell Work

Combine like terms.

1. $(3y + 4.6) + (-5y - 3.25)$

$$= (3y + (-5y)) + (4.6 - 3.25) = -2y + 1.35$$

2. $(6\frac{3}{4}x - 7) + (-\frac{5}{2}x + 7)$

$$= (\frac{27}{4}x + (-\frac{10}{4}x)) + ((-7) + 7)$$

$$= \frac{17}{4}x \text{ or } 4\frac{1}{4}x$$



Consider having students circle or underline like terms before combining. This will provide a visual representation that can aid in understanding of what like terms are.

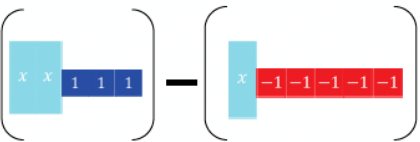
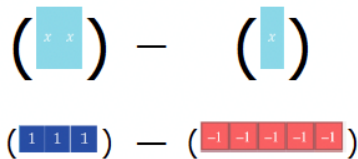
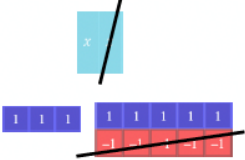
13.9.2 Exploration: Subtracting Expressions with Algebra Tiles

Component Overview: Students use algebra tiles to find the difference of the algebraic expression from the previous lesson's 13.8.3 Collaboration.



13.9.2 Exploration: Subtracting Expressions with Algebra Tiles

- Algebra tile representations are given below. Use the algebra tiles to find the difference.

On the lines, write the algebraic expression for each representation.	 $(2x + 3) - (x - 5)$
Rearrange the tiles to show subtraction of like tiles together. Draw the representation.	
If necessary, create any zero pairs needed for subtraction.	
Find the difference by subtracting (taking away) like tiles. Write the algebraic representation of the difference.	 $x + 8$



Make note of which students recognize that both x and -5 are being subtracted from $2x + 3$ compared to students who only subtract x and then add -5 . Instead of formalizing and explaining the misconceptions, use this activity as an opportunity to instead notice the difference when comparing with peers.



- Some students found the difference to be $x - 2$ while others found it to be $x + 8$. Why do you think that is? Which is correct?
- What is being subtracted from $2x + 3$?

13.9.3 Collaboration: Subtracting Expressions

Component Overview: Students work with a partner to evaluate the given work of four students. Read through each student's statement and work as a class, then partner students together to begin their evaluation. Make sure students not only find the error but also describe how to correct it.



13.9.3 Collaboration: Subtracting Expressions

1. Noah, Lin, Jada, and Andre are rewriting the expression $(x + 8) - (4 - 9x)$ as an equivalent expression with fewer terms. Each student described their work.

Discuss with your partner which students made an error. Describe each student's error and correct the error.

- a. Noah says, "I dropped the parentheses, combined like terms, and ended up with $-8x + 4$."

$$\begin{aligned}(x + 8) - (4 - 9x) \\ x + 8 - 4 - 9x \\ -8x + 4\end{aligned}$$

Noah did not distribute (-1) to both terms. After distributing (-1) to both terms, the result is $x + 8 - 4 + 9x = (x + 9x) + (8 - 4) = 10x + 4$.

- b. Lin says, "I simplified inside the parentheses and ended up with $13x$."

$$\begin{aligned}(x + 8) - (4 - 9x) \\ (8x) - (-5x) \\ 8x + 5x \\ 13x\end{aligned}$$

Lin combined terms that were not like terms. Lin should group the like terms together $(x + 9x) + (8 - 4)$ and then combine like terms to obtain $10x + 4$.

- c. Jada says, "I know that subtracting is adding the opposite, and ended up with $4 + 10x$."

$$\begin{aligned}(x + 8) - (4 - 9x) \\ (x + 8) + (-4 + 9x) \\ x + 8 - 4 + 9x \\ 4 + 10x\end{aligned}$$

Jada has a correct answer. The explanation should include intermediate steps.



This activity acts as a primer for 13.9.4 Guided Instruction. Monitor student responses while they collaborate and make note of any responses that you can use to build on when formalizing approaches in 13.9.4. Students should analyze and compare student responses before writing what students may have done incorrectly, as seeing the different examples will help them solidify their initial thoughts.



Consider providing sentence stems for the written explanations, such as:

- "Noah's answer is correct/incorrect. I know this because..."
- "The mistake he/she made was..."
- "It can be corrected by..."

This is a Collaboration, so prompt students to refine their responses with their partners.

13.9.3 Collaboration: Subtracting Expressions (continued)

- d. Andre says, "I used the distributive property since there is an invisible (-1) in front of the second parentheses to distribute, but I ended up with $10x + 4$."

$$\begin{aligned}(x + 8) - (4 - 9x) \\(x + 8) - 1(4 - 9x) \\x + 8 - 4 + 9x \\10x + 4\end{aligned}$$

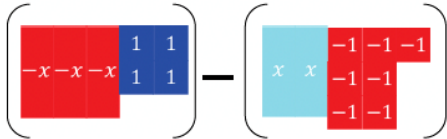
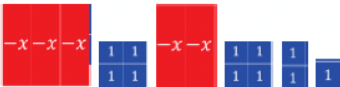
Andre has a correct explanation and answer.



Consider probing questions for students or partners that are struggling to identify the differences in student work.

- Compare Noah and Jada's work. What is the same? What is different?
- Compare Jada's work to Andre's work. What is the same? What is different?

2. When subtracting expressions, use the distributive property to add the opposite. With the algebra tile representations below, model finding the difference by adding the opposite.

On the lines, write the algebraic expression for each representation.	 $(-3x + 4) - (2x - 7)$
Rewrite subtracting to adding the opposite by changing (or flipping) the sign of the 2 nd expression. Draw the representation.	 $(-3x + 4) + (-2x + 7)$
Combine any zero pairs to find the sum. Write the sum.	$-5x + 11$

13.9.4 Guided Instruction: Subtracting Linear Expressions

Component Overview: Review the statements as a class, then the teacher models how to use both methods to complete the table. Have students engage in a Notice and Wonder routine before answering the written response question.



13.9.4 Guided Instruction: Subtracting Linear Expressions

- To subtract linear expressions,
 - add the *opposite*, or additive inverse of each term in the expression.

OR

- distribute a *negative one* to each term in the expression.
- Complete the table below to show both methods.

Additive Inverse Method	Applying the Distributive Property
$(4x + 5) - (-2x + 6)$ $= (4x + 5) + (2x + (-6))$ $= (4x + 2x) + (5 + (-6))$ $= 6x + (-1)$ $= 6x - 1$	$(4x + 5) - (-2x + 6)$ $= 4x + 5 + 2x - 6$ $= (4x + 2x) + (5 + (-6))$ $= 6x + (-1)$ $= 6x - 1$

- When adding and subtracting, parentheses are used to group expressions. Why is the use of parentheses especially important when subtracting?

The use of parentheses ensures that the entire expression is subtracted. This means the negative 1 is distributed to all terms in the expression.



Consider using the following questions to extend student thinking and formalize their understanding of each approach following their written responses:

- How is the expression $(4x + 5) - (-2x + 6)$ different from the expression $4x + 5 - -2x + 6$?
- What do the parentheses communicate about what is being subtracted?

13.9.5 Guided Practice: Subtracting Linear Expressions with Rational Coefficients

Component Overview: Students find the differences in linear expressions. Students work with their partners after seeing the teacher model strategies in the Guided Instruction. If, after the Guided Instruction, it appears students need an additional example, considering working through one of the Guided Practice problems as a whole class before releasing to partners.



13.9.5 Guided Practice: Subtracting Linear Expressions with Rational Coefficients

1. Find the difference of each expression.

a. $(3y + 3.5) - (8.2y - 6)$ $= (3y + 3.5) - 8.2y + 6$ $= (3y + (-8.2y)) + (3.5 + 6)$ $= -5.2y + 9.5$	b. $(m - \frac{3}{8}) - (3\frac{1}{4} - m)$ $= (m - \frac{3}{8}) - 3\frac{1}{4} + m$ $= (m + m) + ((-\frac{3}{8}) + (-3\frac{1}{4}))$ $= 2m - 3\frac{5}{8}$
c. $(-2 + \frac{9}{2}y) - (5.3y - 5\frac{1}{2})$ $= (-2 + \frac{9}{2}y) - 5.3y + 5\frac{1}{2}$ $= (\frac{9}{2}y + (-5.3y)) + (-2 + 5\frac{1}{2})$ $= -0.8y + 3.5 \text{ or } -\frac{4}{5}y + 3\frac{1}{2}$	d. $(-7.5t - 3) - (7 + 3\frac{1}{8}t)$ $= (-7.5t - 3) - 7 - 3\frac{1}{8}t$ $= -7.5t - 3\frac{1}{8}t + (-3 + -7)$ $= -7.5t - 3\frac{1}{8}t + (-10)$ $= -10\frac{5}{8}t - 10$

2. What is the result when $-3x + 4.48$ is subtracted from $-2.2x + 3$?

- Write the algebraic expression.
 $(-2.2x + 3) - (-3x + 4.48)$
- Rewrite the expression without parentheses.
 $-2.2x + 3x + 3 + (-4.48)$
- Simplify the expression by rewriting it using the fewest terms possible.
 $0.8x - 1.48$



Consider pulling a small group for students who would benefit from a model or think-aloud that highlights the usefulness of explicitly writing the coefficient -1 for better visual scaffolding of the distribution.



Notice the phrase “subtracted from” in question 2. Students might invert the order. Consider providing this prompt as a self-check for students:

I see some students found the difference to be $0.8x - 1.48$ while others found it to be $-0.8x - 1.48$.

- Which is correct?
- Which expression is being subtracted?

13.9.6 Collaboration: Adding and Subtracting Linear Expressions

Component Overview: Students work with a partner to analyze, discuss, and correct students' errors in three linear expressions with integer coefficients. Circulate as students complete the problem together and make note of common misconceptions to address whole class with. Review student responses as a class and clarify misconceptions noted during circulation.



13.9.6 Collaboration: Adding and Subtracting Linear Expressions

1. Two students made errors when rewriting the expression $(6x + 9) - (7 - x) + (8x + 11)$.

Zoe's Work	Patricia's Work
$(6x + 9) - (7 - x) + (8x + 11)$	$(6x + 9) - (7 - x) + (8x + 11)$
$(6x + 9) - (7x + 18)$	$(5x + 2) + (8x + 11)$
$-x - 9$	$13x + 13$

- a. Working with your partner, write an explanation to Zoe that describes her error.

Zoe, it looks like you did not follow the order of operations or recognize that a -1 was being distributed to $(7 - x)$. You subtracted x from $8x$ and added 7 and 11 , but you can't combine these terms until you have first distributed the negative to $(7 - x)$. Once that is done, then you can combine like terms from left to right.

- b. Write an explanation to Patricia that describes her error.

Patricia, it looks like you did not distribute -1 to both terms in the expression $(7 - x)$. You subtracted 7 from 9 , but you should have added x to $6x$ rather than subtracting it because subtracting $-x$ is the same as adding x .

- c. Simplify the expression $(6x + 9) - (7 - x) + (8x + 11)$.
 $= 6x + 9 - 7 + x + 8x + 11$
 $= 15x + 13$



Consider providing an additional entry point for verbal processors by intentionally scaffolding the error analysis as a Think-Pair-Share in which students first analyze independently before sharing responses with a partner for more precise responses.



Following question 1b, give students an opportunity to formalize their learning before applying in question 1c.

- When have you made this error before?
- What do you look for to avoid this error in the future?
- What will that look like in question 1c?

13.9.7 Your Turn!

Component Overview: Students work independently to combine like terms and simplify three linear expressions with rational coefficients.



13.9.7 Your Turn!

1. Rewrite each expression by combining like terms.

$$\begin{aligned} \text{a. } & (1 - 4y) - (2.5y + 7) - (-0.3y + 3) \\ & = (-4y - 2.5y + 0.3y) + (1 - 7 - 3) \\ & = -6.2y - 9 \end{aligned}$$

$$\begin{aligned} \text{b. } & \left(3\frac{1}{2}x - 4\right) + (x - 3.2) - \left(-\frac{1}{4}x + 9\right) \\ & = \left(3\frac{1}{2}x + x + \frac{1}{4}x\right) + (-4 - 3.2 - 9) \\ & = 4\frac{3}{4}x - 16\frac{1}{5} \text{ or } 4.75x - 16.2 \end{aligned}$$

$$\begin{aligned} \text{c. } & -(2r + 5) - (6 - 3r) + (10r - 4) \\ & = (-2r + 3r + 10r) + (-5 - 6 - 4) \\ & = 11r - 15 \end{aligned}$$



For extension and error analysis practice (MA.K12.MTR.4.1), prompt students to determine what a student did wrong if they got the following answers:

- $-1.2y - 9$ (added $+2.5y$ instead of $-2.5y$)
- $4\frac{3}{4}x + 1.8$ (added $+9$ instead of -9)
- $5r - 15$ (added $-2r - 3r$ instead of $+3r$)

13.9.8 Wrap-Up: Ticket Out the Door

Component Overview: Students combine like terms to simplify two linear expressions with rational coefficients.



13.9.8 Wrap-Up: Ticket Out the Door

Write the following expressions using the fewest possible terms.

1. $(5x - 13\frac{1}{4}) - (4\frac{1}{2} - 5x)$

$$= 5x + 5x - 13\frac{1}{4} - 4\frac{1}{2}$$

$$= 10x - 17\frac{3}{4}$$

2. $(0.7 - 2.4x) + (-5.32 + 4\frac{1}{10}x)$

$$= 0.7 - 5.32 - 2.4x + 4\frac{1}{10}x$$

$$= -4.62 + 1.7x$$



Students may get $0x - 17\frac{3}{4}$ or $-17\frac{3}{4}$ for the first linear expression. Remind students that writing the coefficient of -1 in front of the second parenthesis can help them remember to distribute.



While the lesson's learning target is focused on subtracting linear expressions, this ticket out the door is intended to be used as formative assessment for both adding and subtracting linear expressions following the first three lessons of the unit. Use the informal data gathered from this formative assessment to plan for adjustments or remediation prior to the end of the unit.